

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Electrical contact cleaner, Part Number 3824510
- Fuel System Clean Care Kit, Part Number 4919073.

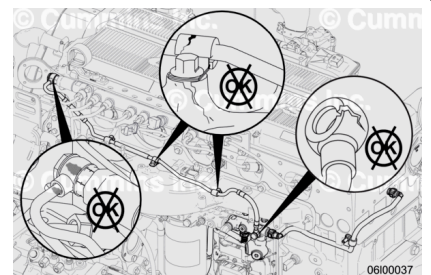
Additional Service Items

- No additional service items required.

Initial Check

Inspect the fuel drain lines for any signs of leaks, cracks, chafing, and loose or broken brackets.

Repair or replace any damaged parts.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.

⚠ WARNING ⚠

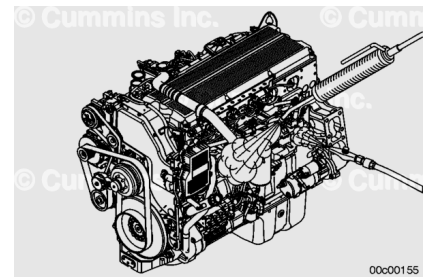
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Clean all fittings before disassembly. Dirt or contaminants can damage the fuel system.



Before servicing **any** fuel system components, (such as fuel lines, fuel pump, injectors, etc.) which would expose the fuel system or internal engine component to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component to be removed. Dirt or contaminants can be introduced into the fuel system and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine. Refer to Procedure 000-009 in Section 0.

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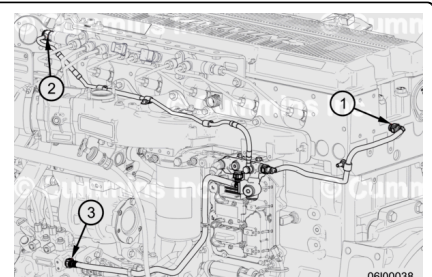
Clean the fuel drain line and mating components with electrical contact cleaner, Part Number 3824510, or equivalent.

To prevent damage from debris and contamination, cover, cap, or plug any openings as soon as possible when servicing the fuel system. Caps and plugs can be found in Fuel System Clean Care Kit, Part Number 4919073.

Remove

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



⚠ WARNING ⚠

Do not vent the fuel system on a hot engine; this can cause fuel to spill onto a hot exhaust manifold, which can cause a fire.

The injector drain connects at the back of the cylinder head (1) with a quick disconnect style check valve and a copper sealing washer.

The copper washer **must** be replaced each time the quick disconnect style check valve is removed.

The fuel rail pressure relief valve (2) connects to the fuel rail with an M16 banjo bolt.

The fuel pump drain (3) connects to the fuel pump head with an M12 banjo bolt.

⚠ CAUTION ⚠

Do not use a screwdriver or pliers on the locking tabs. This may result in damage to the quick disconnect fitting.

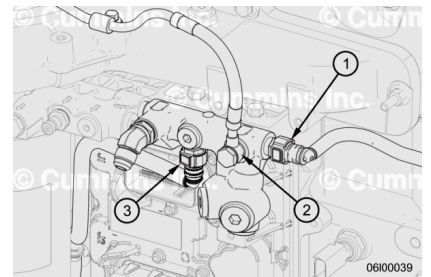
Note : A fuel thermal recirculation valve is installed as part of the fuel drain manifold.

There are two variations to the fuel drain manifold:

1. Drain manifold integrated thermal recirculation valve.
 2. Drain manifold integrated thermal recirculation valve with check valve.
- The injector drain line connects the back of the cylinder head (1) to the thermal recirculation valve with a quick disconnect fitting.
 - The fuel rail pressure relief valve (2) connects to the thermal recirculation valve with an M12 banjo bolt.
 - The fuel pump drain (3) connects to the thermal recirculation valve with a quick disconnect fitting.

Remove the P-clip capscrews, stud, banjo bolts, check valve, and sealing washers.

Remove the fuel drain lines.

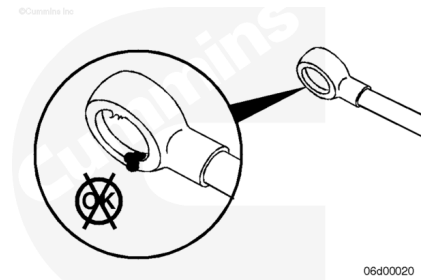


Inspect for Reuse

Inspect the P-clips, drain lines, and banjo fittings for damage.

Inspect the banjo seal washers for damage.

Replace any damaged components.



Install

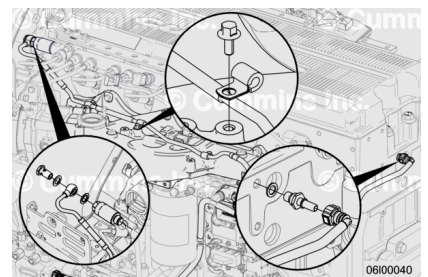
Install the drain lines on the engine.

Loosely install the P-clip capscrews, stud, quick disconnect style check valve, and sealing washers for each drain line.

Always use a new copper washer with quick disconnect style check valve.

Tighten the P-clip capscrews, banjo bolts, and quick disconnect style check valve.

Install quick disconnect fittings, if equipped.



Torque Value:

M12 Banjo Bolts and Check Valve 24 n•m [212 in-lb]

Torque Value:

Quick Disconnect Check Valve with Copper Washer 54 n•m [40 ft-lb]

Torque Value:

M16 Banjo Bolts 41 n•m [30 ft-lb]

Torque Value:

P-Clip Capscrew 20 n•m [177 in-lb]

Torque Value:

P-Clip Stud 47 n•m [35 ft-lb]

Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.